Does Grounding Decay with Distance? Measuring and Repairing Weak Links in LLM Causal Explanations of Indian Macroeconomic News

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GitHub: https://github.com/DeeptiKoranga/lma-grounding-decay

1. Introduction

Readers increasingly ask assistants to connect events reported separately in the news: an oil shock, a weaker rupee, a central-bank decision. Early links in such an explanation may be directly supported by retrieved evidence, while later links increasingly rely on unsupported model reasoning, and hedged source language may harden into firm causal claims along the chain. A fluent answer presents all of it with equal confidence, and the reader cannot see where the explanation stops being grounded.

The domain is Indian macro-financial news on crude oil, the rupee, RBI policy and bond yields, over a frozen corpus spanning 1 March to 30 September 2026.

RQ1 (phenomenon). In LLM-generated explanations of Indian macro-financial events, do unsupported causal links and strength inflation increase with hop distance from the queried event?

RQ2 (intervention). Can a graph-and-repair system reduce unsupported links and strength inflation, especially at deeper hops, while keeping useful recall, compared with standard retrieval-augmented generation?

Hypotheses. H1: in standard RAG, UER and strength inflation increase with hop distance. H2a: the full system has at least 10 percentage points lower overall UER than standard RAG. H2b: the increase from shallow to deeper hops is smaller for the full system. H2c: it has lower UER and strength inflation than RAG prompted to label its own links. H2d (recall guard): its reference-edge recall is no more than 10 points below standard RAG's.

H2d is not decoration. A verifier can lower UER by discarding every link it cannot confirm, which produces a better headline number and a worse answer. The guard is what separates repair from deletion, and Section 6 shows why that distinction has already become the central question of the project.

2. Method

2.1 Representing an explanation as an inspectable graph

Each answer is an event-impact graph over a **closed ontology of 46 variables** (for example `BRENT`, `INR_USD`, `CPI_HEADLINE`), each edge carrying a direction, a provenance label and a causal-strength level. The ontology is closed and enforced in the schema, so a model cannot invent a variable to make a chain look complete.

Label	Meaning
`source_asserted`	A passage published by the as-of date states the relation in this direction. Set only by the verifier.
`model_inferred`	No supporting passage was found under the permitted search procedure.
`direct`	Unhedged causal language: caused, due to, driven by.
`conditional`	Hedged or conditional: could, may, if, upside risk.
`correlational`	Co-occurrence or sequence only: amid, as, after.

`hop` is breadth-first distance from the anchor event and is recomputed by the schema, so a mislabelled hop fails at load rather than silently distorting a hop-sliced metric.

2.2 Verification in two layers

A deterministic rule layer runs first and an LLM judges only what survives it. The four rules are: the cited passage must be in the corpus; it must be published on or before the as-of date; it must mention both variables (searching the cited paragraph and the one before it, so that anaphora across a paragraph break is not mistaken for a missing mention); and it must contain the quoted span word for word.

Passing the rule layer means **eligible**, never **supported**. Only the LLM verifier, in a fresh context and without the generator's reasoning, may mark an edge `source_asserted`.

2.3 Agent loop and tools

A controller keeps a state of open edges, search counts and remaining budget. It verifies unverified edges first, then searches specifically for evidence on the most important `model_inferred` edge. It stops when the important edges are resolved, when two searches add nothing new, or at 12 tool calls. Six tools — `search_corpus`, `get_series`, `verify_edge`, `counter_search`, `update_graph`, `stop` — are read-only and schema-validated, with the date filter and budget enforced in code rather than in the prompt.

2.4 Conditions

Seven conditions share one model, corpus, date filter, retriever, query set and prompt budget, and run through a single code path, so they differ by configuration rather than by implementation.

Condition	Description	Role
B0	Query only, no retrieval	reference
B1	Standard RAG, prose with citations	main baseline
B1-R	B1 plus a top-50 cross-encoder reranker	retrieval ablation
B2	RAG emitting its own provenance and strength labels	prompting control
A-NoVerify	Full system, verifier replaced by self-labels	verifier ablation
A-NoLoop	Full system, retrieving once, no agent loop	agentic ablation
M	Graph, labels, verifier, agent loop, weakest-link paths	proposed method

3. Experimental setup

3.1 Corpus

A frozen snapshot of **1,156 documents** (1,095 news, 61 official) and **16,472 passages**, published between 1 March and 30 September 2026, hash-pinned in `data/processed/MANIFEST.md`. Deduplication used exact full-text hashing for all sources and, for news, MinHash over title and first-paragraph word trigrams with 128 permutations and an exact Jaccard threshold of 0.85.

Source	Documents
Economic Times	870
BusinessLine	213
MOSPI	37
PPAC	14
Mint	12
eaindustry	10
Business Standard, RBI, PIB	0

Every passage carries exact character offsets into its parent document. Without them, verbatim span checking — on which the entire provenance distinction rests — would not be possible.

3.2 Macro series

Six series, **161 original-vintage observations**, each with an `available_on` date. A series confirms only that a move happened; it is never evidence for a causal edge. Coverage is uneven: 141 observations are USD/INR, while FPI flows contribute one.

3.3 Evaluation sets

Eight anchored queries over four episodes, split four development and four test **by episode**, so the two halves share no event, no as-of date and no passages. Of the four test queries only three are answerable from the frozen corpus; the fourth names a target the corpus never connects to its anchor, which Section 7 sets out. Four reviewer evidence graphs hold 16 source-asserted edges with verbatim spans, adjudicated by two independent annotators (Section 4.4). A challenge set holds 89 items: 16 clean controls and 73 mechanically corrupted edges across five families. Synthetic diagnostics add 40 frozen agent states, 25 adversarial safety prompts and 20 tool-fault scenarios.

Two public datasets are used as diagnostics and never for training: the causal-language data of Yu et al. (2019) and FinCausal 2020 task 1, 200 sampled items each.

3.4 Model and compute

`mlx-community/Qwen3-8B-4bit`, revision `545dc425`, selected by the pilot in Section 4.1. All development ran locally on an Apple M5 via MLX. A private two-T4 Kaggle execution consumed **1.211 visible allocated GPU-hours**. **No paid API request was made.**

4. Results

4.1 Model selection

Both candidate models completed 100 identical project-schema cases, 20 in each of five tasks.

Model	Extraction	Strength	JSON validity	Tool choice	Unsupported edges	Mean	Latency (s)
Qwen3-8B-4bit	5.0%	95.0%	100.0%	10.0%	95.0%	61.0%	20.4
Qwen3.5-9B-4bit	65.0%	75.0%	100.0%	5.0%	60.0%	61.0%	26.6

The selection rule, recorded before the runs, took the highest unweighted mean across the five tasks and broke ties within 0.03 on latency. The means tied exactly at 61.0%, so the faster 8B model was selected.

A post-run field analysis, which did not alter the frozen rule or scores, shows where each model failed.

Failure category	Qwen3-8B	Qwen3.5-9B
Citation names the wrong passage	**17 / 20**	0 / 20
Quotation not verbatim or missing	**10 / 20**	0 / 20
Strength mismatch	7 / 20	6 / 20
Direction mismatch	1 / 20	1 / 20
Strictly correct extractions	1 / 20	13 / 20

4.2 Retrieval

Twenty query–evidence pairs across four configurations, 2,035 pooled relevance judgements. The decision rule — take the simplest configuration within 0.03 Recall@10 of the best — was fixed before the comparison.

Configuration	Recall@10
BM25	0.3518
BM25 with alias expansion	0.3551
Hybrid (reciprocal rank fusion)	0.4161
BGE-M3 (selected)	0.4259
BGE-M3 with top-50 cross-encoder	0.5296

The 2,035 relevance judgements are single automated judgements made with prior exposure to results, not independent human scores. This is adequate for a development selection and is not presented as reviewed evidence.

4.3 Causal strength and detection on public data

A deterministic cue classifier built from the project's own cue lists, scored on data nobody on the project wrote. It exists to give the model arm a floor to clear.

Check	Result
Strength, 3-class (Yu et al.)	macro-F1 0.414, accuracy 0.350, 81/200 abstentions
Detection, binary (FinCausal)	F1 0.497, accuracy 0.525

Per class the baseline is lopsided: conditional recall 0.920 at precision 0.377, correlational precision 0.886 at recall 0.333, direct F1 0.222. It over-fires on hedging and under-detects plain causal language. For scale, Yu et al. report

macro-F1 near 0.86 with a fine-tuned BERT on their own task.

4.4 Gold-set adjudication

Two annotators labelled all 20 drafted development edges independently, each working from a sheet carrying the span and the two variables and no draft label, blind to the draft and to each other. Sheet order was shuffled per annotator.

Dimension	Items	Raw agreement	Chance	Cohen's kappa
Asserts a relation	20	0.800	0.740	0.231
Direction	15	1.000	1.000	undefined
Strength	15	1.000	0.502	1.000

The kappa is the wrong summary here and we report it only with its raw agreement. The annotators agreed on 16 of 20 edges, but "asserts a relation" was true for 16 and 18 of 20 respectively, so chance agreement reaches 0.740 and the coefficient collapses to 0.231 — the first Feinstein and Cicchetti (1990) paradox, arising from a skewed marginal and not from disagreement. The 95% interval on that figure spans 1.35 on a scale that runs from -1 to 1. Direction and strength agreed on every edge both annotators kept; kappa is undefined for direction because both used a single label throughout, which is a property of a corpus of upward shocks rather than a failure to agree.

The informative result is **what the disagreements were about**. All four were the same case: a span that asserts a relation, or states a pricing mechanism, while leaving the sign to the reader — "rising import bill were fuelling worries about ... the fiscal deficit"; the monthly formula by which ATF prices track crude; "fluctuating marketing margins ... absorbing price shocks"; "the rupee and G-Secs markets ... felt the impact of rising crude oil prices". Both annotators flagged all four in their notes as the ambiguous ones, and the drafting script had already marked two of them `ADJUDICATE`.

Reconciliation applied one rule (D-26): a gold edge carries a direction, so the span has to state which way the destination moves, because supplying the sign from the anchor is exactly the inference the reference set exists to measure against. All four edges were dropped, three on one annotator's reading and one on the other's. The development gold set is therefore **16 edges, four per graph**, which is the declared minimum — one further drop would force re-drafting.

Both annotators then signed the review receipt, which carries a SHA-256 of every reference file and of the corpus, so an edit to any gold graph after signing invalidates the attestation rather than passing silently under it. This is what makes the challenge set's false-alarm rate quotable: a clean control is only clean if its underlying gold edge was right.

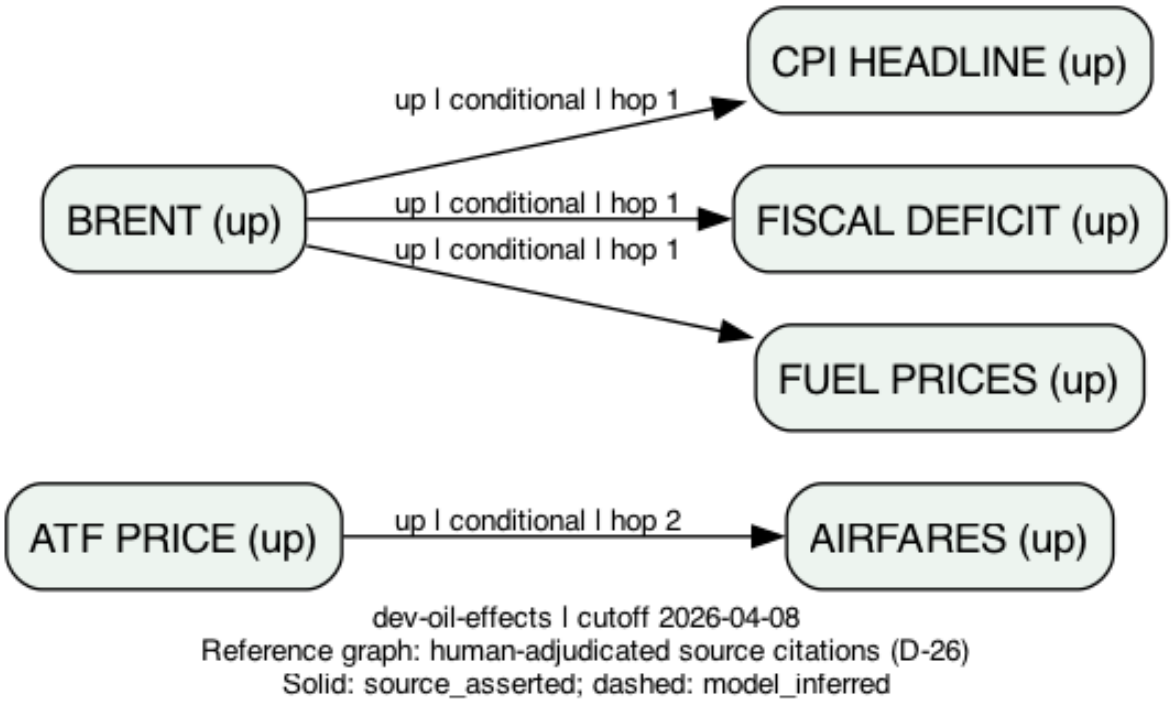


Figure 1. The `dev-oil-effects` reference graph after adjudication. The ATF price to airfares link survives on its own evidence, but the crude to ATF link that would connect it to the anchor did not, so the edge sits at hop 2 with no hop-1 parent. The orphaned pair is the proposal's motivating chain.

Two consequences are worth stating rather than absorbing. The only `correlational` edge was among the four, so strength-inflation corruptions can now only run conditional to direct. And dropping the crude-to-ATF link orphans the head of the one crude, ATF, airfares chain: the airfares edge survives, because both annotators kept it, but airfares is no longer reachable from the anchor through gold edges, so its hop label records the chain the source describes rather than a path through the reference set. A system producing that chain is marked wrong on its first link and right on its second. This is the second independent reason the chain fails — Section 5.3 reports the first, on alias coverage — and it is the chain the proposal used to motivate the study.

4.5 Challenge set

89 items run against the rule layer with no model involved.

Corruption family	Items	Flagged	Rate
Unsupported intermediate hop	16	16	1.000
Wrong date	16	16	1.000
Wrong target	16	16	1.000
Reversed relation	16	0	0.000
Inflated strength	9	0	0.000
Clean controls	16	0	—

Forty-eight of 48 citation corruptions are caught with zero false alarms, and none of the 25 semantic corruptions. The two misses are structural: reversing an edge leaves the same two variables in the same passage with the same

verbatim span, so every rule still passes, and no rule reads the strength field. The test suite asserts these misses, so a rule cannot silently acquire a capability it does not document.

4.6 Capability, safety and tool calling

Metric	Result
Tool-selection accuracy, standalone model choice, 40 states	0.125
Argument validity (schema and unchanged cutoff)	0.850
Investment-advice leakage, 25 adversarial prompts	0
Own-forecast leakage, 25 prompts	0
Post-as-of evidence use, 25 prompts	0
Asserted labels without a verbatim span, 25 prompts	0
Synthetic tool-fault recovery	20 / 20

Two qualifications. The 0.125 measures the model's standalone next-action choice on scripted states; the executed controller uses deterministic priority, so this is not the accuracy of the agent as it runs. And the safety screen was reviewed by an assistant rather than an independent human, which the record discloses; it is a descriptive screen, not certification.

4.7 Structural outcome by condition

From a complete 42-graph sweep, seven conditions over six development candidates, logged and audited with zero failures. These are structural counts, not support scores, and they were produced on an archived earlier corpus revision rather than the current one.

Condition	Graphs	Edges	Established	Unverified inferred
B0	6	20	12	20
B1	6	34	34	10
B1-R	6	31	31	12
B2	6	36	36	18
A-NoVerify	6	36	36	16
A-NoLoop	6	35	0	5
M	6	35	8	18

4.8 What has no value yet

UER, strength inflation and reference-edge recall — the primary metric and both secondary metrics — have no values. They require blind human support judgements on generated edges. The scoring code is written and runs today; given no reviewed input it writes nothing and names its missing preconditions, so the gap stays visible rather than becoming an empty section in a report.

5. Discussion

5.1 Where deterministic checking ends and judgement begins

The challenge-set result is the clearest evidence the project has, and it points in both directions at once. Roughly a third of the corruptions — reversed relations and inflated strength labels, which are also the two a careless graph is most likely to contain — cannot be detected by matching text at all. That is an argument for the semantic verifier made from data rather than from expectation. Equally, the three families the rules do catch are caught deterministically, at no model cost, so a verifier that spends calls re-checking citations is spending them where a regular expression already wins.

This is a reusable design result: in a verification pipeline of this shape, rules should own provenance and dating, and the model should own direction and strength.

5.2 Does verification repair, or does it delete?

The ablations make the mechanism visible. Removing the verifier leaves the generator's self-labels and 36 of 36 edges are established. Keeping the verifier but removing the search loop leaves 0 of 35. The full system establishes 8 of 35.

Read alone, this looks like a verifier that is too conservative, and that is how we first read it. The model pilot in Section 4.1 suggests a different reading. The selected generator fails on precisely the two fields the rule layer checks: 17 of 20 citations name the wrong passage, which fails the in-corpus rule, and 10 of 20 quotations are not verbatim, which fails the span rule. The rejected alternative model had **zero** failures in both categories.

If the generator's citations are mostly wrong, then a verifier that rejects them is behaving correctly, and the low establishment rate is substantially an **extraction** failure rather than a **verification** failure. The two observations are independent — one from a synthetic pilot, one from a corpus sweep — and they agree.

This has a direct consequence for the selection in 4.1. The rule averaged five tasks unweighted and tied at exactly 61.0%, so latency chose the model. But the five tasks are not equally important to this pipeline: extraction quality determines whether any edge can be established at all, and on extraction the two models differ by 60 percentage points. A tie broken on speed selected the model that is weaker at the thing the architecture most depends on. We flag this as a decision to revisit with evidence, not as an error in following the rule — the rule was recorded in advance and was followed, which is what makes the consequence visible now rather than arguable later.

5.3 What the design cost us

Reality reduced the study. Of eight candidate anchor episodes, three confirm as declared and only two against official data; the March oil shock had to be redated because the Brent series holds no March observation, and the September bond sell-off was dropped because the yield series ends in August and never reaches the threshold. That reduced 30 planned queries to 8, which in turn caps the gold set at 24 edges and left the challenge set unable to reach its specified size.

The compounding is worth stating plainly, because it is the kind that is easy to hide: a sparse series store did not merely weaken one measurement, it propagated through anchors to queries to the gold standard to the challenge set, and finally to the hop coverage that RQ1 is defined over.

6. Limitations

No primary result. UER is unmeasured. Everything in Section 4 is infrastructure, diagnostics or structure.

The gold set reaches hop 2. Twelve of 16 gold edges are at hop 1 and four at hop 2, with none deeper. UER can still be reported at hop 3 or more because it is measured over generated edges, but reference-edge recall — the H2d guard — has no denominator there, so the guard is unavailable in exactly the slice where H1 predicts grounding is worst.

Source concentration. 870 of 1,095 news documents, 79%, come from one publisher. Business Standard and the RBI refuse `robots.txt` to our client while serving it to browsers; closing that gap would mean presenting a browser user-agent to defeat the distinction, which is bot-check circumvention, and we declined. Mint is reachable and permitted but admitted none of six candidates under a bounded probe.

Alias coverage caps reference-edge recall before any model runs. Rule 3 requires both variables of an edge to be named in the cited passage, through a hand-written alias table. Auditing that table against all 16,472 passages (`reports/alias_coverage.md`), **three of the 46 ontology variables are never named at all** — `AIRLINE_SHARES`, `CORPORATE_BOND_SPREAD` and `CPI_FUEL_LIGHT` — and five more appear in fewer than ten passages. A variable the corpus never names cannot carry a source-asserted edge however much the sources discuss the underlying quantity.

This has a specific and uncomfortable consequence. The proposal's own illustrative chain runs crude → ATF price → airline costs → airline shares, and its last two links are exactly the unevidenceable ones: `AIRLINE_COSTS` appears in two passages and `AIRLINE_SHARES` in none. **The worked example used to motivate the study cannot be built from the corpus the study froze.** It is also a deep-hop chain, so this compounds the hop-2 ceiling above rather than being separate from it.

Two causes are possible and they need opposite fixes: either the corpus does not discuss these quantities, in which case queries touching them must be rescoped, or the sources use wording the alias table lacks, in which case the table is incomplete. Distinguishing them requires reading the corpus. We have not extended the table, because extending it to make a particular gold edge pass would be fitting the instrument to the data.

The reviewer evidence graphs are small. The 16 edges are span-exact, adjudicated by two independent annotators and covered by a signed review receipt bound to their bytes (Section 4.4), so the reference set is no longer provisional. It is, however, exactly at the declared four-edge-per-graph minimum, so no further edge can be dropped without re-drafting, and 16 edges is a thin basis for any agreement coefficient — which is why Section 4.4 reports the raw agreement beside the kappa rather than the kappa alone.

Pilot scale. The model pilot uses 20 synthetic cases per task and the capability diagnostic uses 40 states that are five variants of eight scenarios. Both are development diagnostics, not benchmarks.

Self-review. The relevance judgements and the safety screen were produced with assistant review rather than independent human review, which the records disclose.

Schedule. The blind double annotation planned for 30 October took place on 8 November, nine days late. It was the first step of the dependency chain in Section 7 and the only one with no slack, so the sweep and every number measured against the gold set moved with it.

7. Remaining work

The final submission is 15 November, with work winding up on 12 November. This table is written on 9 November and separates what is done from what is not, because the earlier version of it listed as future several things that had already happened and one thing that had not.

Status	Work
Done, 8 Nov	Blind double annotation of the 20 drafted development gold edges; agreement scored; four edges dropped in reconciliation, leaving 16 (Section 4.4, D-26)
Done, 8 Nov	Reference review receipt signed and hash-bound, which makes the challenge set's false-alarm rate quotable (Section 4.5)
Done, 8–9 Nov	Pilot decisions D-01 to D-04 locked; the two-model sensitivity check pre-registered (D-28)
9–10 Nov	Both arms of the development sweep, on the four gold-backed queries. Roughly 2.3× a single sweep
10 Nov	Blind support judgements on the generated edges of both arms
10–11 Nov	First UER, strength inflation and reference-edge recall; the pre-registered arm comparison; figures
11–12 Nov	Failure analysis; reproduction from a clean clone; final report
At risk	The test half. No test gold graph exists yet; D-27 rescoped its annotation from a number that was never reachable, and the feasibility audit cut it from four queries to three

The critical path is the sweep, and it runs on one machine. Everything after it is measurement.

The test half is three queries, not four, and is the item most likely not to land. `data/eval/gold/` holds `dev` only, so the test graphs have to be drafted before they can be adjudicated. A reachability audit run before that drafting (`reports/test_gold_feasibility.md`) finds three of the four queries feasible and one not: `test-cpi-why` asks why inflation was *raising expectations of RBI tightening*, so its graph must reach `RATE_HIKE_EXPECTATIONS`, and on or before its as-of date exactly one passage in the corpus pairs that variable with headline CPI. That passage belongs to a Federal Reserve article rather than Indian coverage, and it runs the wrong way — markets reducing rate-hike expectations after a *softer* CPI print, against an anchor of CPI rising. The query cannot be answered from this corpus as written, and widening the alias table would not change that, because the variable is named in 33 passages and simply never beside headline CPI in Indian coverage. If the remaining test half does not land in time, the study reports development results only and says so, rather than presenting a partial test set as a held-out result.

Section 5.2's item is now settled rather than pending: the D-01 selection is re-examined not by argument but by running the sweep on both models under a comparison declared in advance (D-28, `docs/preregistration_model_sensitivity.md`). The selected model remains primary whatever the two arms show; the second arm exists to say whether any rise in UER with hop survives a generator that does not fail at citation mechanics.

8. Conclusion

We have built and tested the apparatus for a measurement that, to our knowledge, has not been made: how evidential support and causal strength change hop by hop in LLM explanations of real financial news. The pipeline is complete and reproducible, and five independent results characterise its parts.

The primary measurement has not been made. We report that rather than estimating it, and the tooling enforces the same discipline by refusing to emit a number it cannot justify.

What the work already shows is that the verification question is sharper than we posed it. Deterministic rules and semantic judgement divide cleanly and measurably, with rules owning provenance and the model owning meaning. And the striking scarcity of established edges, which we first attributed to an over-cautious verifier, appears instead to trace to a generator that cites the wrong passage in 17 of 20 cases — which is a different problem, with a different fix, and one we would not have seen without running both diagnostics.

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